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Electric motor that can be downsized in a rotation axis direction

Abstract

There is provided an electric motor that can be downsized in a rotation axis direction and can also suppress the deformation of a bearing and a bearing house portion along with the deformation of a resin shell. The electric motor includes: a columnar rotor; a shaft disposed along a rotation axis of the rotor; a cylindrical stator core disposed on an outer circumferential side of the rotor; a resin shell that covers the stator core; a bearing that rotatably supports the shaft; and a bearing house portion that stores the bearing. The resin shell includes an annular portion integral with the stator core, and an end surface portion connected to an end portion of the annular portion and expanding from the annular portion to an inner circumferential side. The end surface portion includes a connection portion to be connected to the bearing house portion. The bearing house portion includes a tubular portion and a flange portion extending from an end of the tubular portion to the outer circumferential side in the radial direction. An outer circumferential edge portion of the flange portion is fixed to the connection portion of the resin shell.

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Background/Summary

CROSS REFERENCE TO PRIOR APPLICATION

(1) This application is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2021/012251 (filed on Mar. 24, 2021) under 35 U.S.C. § 371, which claims priority to Japanese Patent Application No. 2020-063645 (filed on Mar. 31, 2020), which are all hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to an inner-rotor electric motor including a rotor disposed coaxially with a cylindrical stator on the inner diameter side of the stator.

BACKGROUND ART

(3) As an electric motor, an inner-rotor electric motor has been conventionally known, in which a columnar rotor including a permanent magnet is disposed coaxially with a cylindrical stator, which generates a rotating magnetic field, on the inner diameter side of the cylindrical stator. This electric motor is, for example, used to rotationally drive a blower fan mounted in an air conditioner.

(4) In the electric motor of Patent Literature 1 as this type of electric motor, a bearing house portion (bearing bracket) that holds a bearing is disposed close to the rotor in the axis direction of the stator, so that the electric motor is downsized in the axis direction. Further, the stator core is covered with a molded resin to form a resin shell.

CITATION LIST

Patent Literature

(5) Patent Literature 1: Japanese Patent Application Laid-open No. 2011-109861

DISCLOSURE OF INVENTION

Technical Problem

(6) Here, in the electric motor of Patent Literature 1, the resin shell that covers the stator covers the bearing house portion as well. In general, the resin is more likely to expand and contract due to heat than metal, and if the resin shell covering the bearing house portion is deformed due to heat, the bearing house portion or the bearing is deformed along with the deformation of the resin shell, which may reduce the slidability of the bearing.

(7) In this regard, it is an object of the present invention to provide an electric motor that can be downsized in an axis direction of a rotation axis and can suppress the deformation of a bearing and a bearing house portion along with the deformation of a resin shell.

Solution to Problem

(8) According to an aspect of the present invention, there is provided an electric motor including: a columnar rotor; a shaft disposed along a rotation axis of the rotor; a cylindrical stator core disposed on an outer circumferential side of the rotor; a resin shell that covers the stator core; a bearing that rotatably supports the shaft; and a bearing house portion that stores the bearing. The resin shell includes an annular portion integral with the stator core, and an end surface portion connected to an end portion of the annular portion and expanding from the annular portion to an inner circumferential side. The end surface portion includes a connection portion to be connected to the bearing house portion. The bearing house portion includes a tubular portion and a flange portion extending from an end of the tubular portion to the outer circumferential side. An outer circumferential edge portion of the flange portion is fixed to the connection portion of the resin shell.

(9) According to the present invention, it is possible to downsize an electric motor in an axis direction of a rotation axis and to suppress the deformation of a bearing and a bearing house portion along with the deformation of a resin shell.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is an overall perspective view of a permanent magnet electric motor according to the present invention.
 - (2) FIG. 2 is a transverse cross-sectional view of the permanent magnet electric motor according to the present invention.
 - (3) FIG. 3 is a perspective view of a bracket of the permanent magnet electric motor according to the present invention.
 - (4) FIG. 4 is an overall perspective view of the permanent magnet electric motor according to the present invention, showing a state in which the bracket of FIG. 3 is removed.
 - (5) FIG. 5 is a cross-sectional view of a cross-section taken along a slitted groove shown in FIG. 1.
- ## MODE(S) FOR CARRYING OUT THE INVENTION

(6) Next, an embodiment of the present invention will be described with reference to the drawings. In the following description about the drawings, the same or similar portions will be denoted by the same or similar reference symbols. It should be noted that the drawings are schematic and may differ from reality. Therefore, specific constituent parts should be determined by referring to the following description.

(7) Further, the embodiment to be described below exemplifies apparatuses and methods for embodying the technical idea of the present invention, and the technical idea of the present invention does not specify the shape, structure, arrangement, and the like of the constituent parts to those described below. Various modifications can be made to the technical idea of the present invention within the technical scope defined by the claims.

(8) Hereinafter, an electric motor according to an embodiment of the present invention will be described.

(9) <Overall Configuration of Electric Motor>

(10) FIGS. 1 to 5 are views for describing a configuration of an electric motor 1 of this embodiment. As shown in those figures, the electric motor 1 is a brushless DC motor, for example. The electric motor 1 is, for example, used to rotationally drive a blower fan mounted in an outdoor unit of an air conditioner, though not shown in the figures.

(11) As shown in FIGS. 1 and 2, the electric motor 1 of this embodiment includes a stator 2, a rotor 3, a motor shell (casing, main body) 10, and a bracket 41.

(12) Hereinafter, an inner-rotor permanent magnet electric motor 1 will be described as an example, in which a columnar rotor 3 including a permanent magnet portion 31 is rotatably disposed inward (inner circumferential side) in the radial direction of a cylindrical stator 2 that generates a rotating magnetic field.

(13) <Rotor, Stator, and Motor Shell>

(14) As shown in FIG. 2, the rotor 3 includes an annular permanent magnet portion 31 and a coupling portion 35, which is disposed on the inner diameter side relative to the permanent magnet portion 31 and couples the permanent magnet portion 31 and a shaft 32 to each other. The shaft 32 is fixed to the columnar rotor 3 along the center axis of the rotor 3. In this embodiment, the permanent magnet portion 31 and the coupling portion 35 of the rotor 3 are formed by integral molding of a resin material in which a ferrite magnetic substance is mixed. After the molding, only the permanent magnet portion 31 is magnetized to cause the permanent magnet portion 31 to function as a ferrite bonded magnet. Further, the permanent magnet portion 31 is magnetized to be a polar anisotropic magnet in which a south pole and a north pole alternately appear in the circumferential direction thereof. Thus, a part of a yoke for concentrating the flow of the magnetic flux of the permanent magnet portion 31 becomes unnecessary, and the leakage flux can be suppressed.

(15) Note that the permanent magnet portion 31 and the coupling portion 35 may be formed separately. For example, the rotor 3 may be a so-called surface magnet (SPM) rotor, in which a

plurality of ferrite sintered magnets (corresponding to the permanent magnet portion **31**), which are obtained by sintering a powder-like ferrite magnetic substance in a mold, are annularly attached to the outer circumferential surface of a rotor core (corresponding to the coupling portion **35**).

(16) The stator **2** includes a stator core **21** including a cylindrical yoke portion (not shown) and a plurality of teeth portions (not shown) extending from the yoke portion to the inner diameter side, and winding (not shown) wound around the teeth portion via an insulator. The stator core **21** is formed of, for example, silicon steel of a soft magnetic material. The stator **2** is covered with the motor shell **10** formed of resin by resin integral molding, except for the inner circumferential surface of the stator core **21** (see FIGS. **2** and **4**). Specifically, the motor shell **10** covers the stator **2** including the stator core **21** and the winding. As shown in FIGS. **1** and **2**, the stator **2** is disposed on the outer circumferential side of the rotor **3** (outward in the radial direction of the electric motor **1**). Further, the stator core **21** of the stator **2** is disposed such that the teeth portion of the stator core **21** faces the permanent magnet portion **31** of the rotor **3** in the radial direction of the stator **2**. In other words, the stator **2** is disposed such that the annular permanent magnet portion **31** of the rotor **3** faces the stator core **21** of the stator **2** in the radial direction.

(17) The motor shell **10** may have any shape. For example, the motor shell **10** may be formed into a hollow cylindrical shape having an open end surface on one side (in the embodiment, the opposite output side of the shaft **32**) in the axis direction of the center axis of the electric motor **1**, that is, the rotation axis of the rotor **3** (hereinafter, rotation axis C). In this embodiment, the motor shell **10** includes an annular portion **12** and an end surface portion **13** formed at the end portion of the annular portion **12** on the opposite side of the opening. The end surface portion **13** expands inward in the radial direction (inner diameter direction) from the annular portion.

(18) The rotor **3** is rotatably disposed on the inner circumferential side of the stator core **21** of the stator **2** with a predetermined clearance (gap) from the stator core **21**. As shown in FIGS. **2**, **4**, and **5**, the permanent magnet portion **31** is annularly disposed on the outer side (outer circumferential side) in the radial direction of the rotor **3** so as to face the stator core **21**.

(19) The rotor **3** is fixed to the circumference of the shaft **32**. The shaft **32** is rotatably supported (held) by a first bearing **33** and a second bearing **34** fixed to the outer circumferential surface of the shaft **32**. Further, the first bearing **33** is stored (held) in a first bearing storing portion **42** (bearing house portion) to be described later, and the second bearing **34** is stored (held) in a second bearing storing portion **43** (bearing house portion) to be described later, so that the rotor **3** is rotatably supported. The first bearing storing portion **42** and the second bearing storing portion **43** are formed of, for example, a magnetic substance of chromium-nickel-based stainless steel.

(20) <Bearing, Bracket, and Bearing House Portion>

(21) As shown in FIGS. **2**, **4**, and **5**, the first bearing **33** is fixed to one end side (opposite output side) of the shaft **32** at the inner race side of the first bearing **33**. The second bearing **34** is fixed to the other end side (output side) of the shaft **32** at the inner race side of the second bearing **34**. The first bearing **33** and the second bearing **34** (a pair of bearings) cooperate to rotatably support (pivotally support) the shaft **32** and the rotor **3** fixed to the shaft **32**. For example, a ball bearing is used for each of the first bearing **33** and the second bearing **34**.

(22) The bracket **41** includes the first bearing storing portion **42** that is formed of a magnetic substance and stores the first bearing **33**, and a non-magnetic portion **44** (end surface portion) formed of a non-magnetic substance (e.g., resin). In the motor shell **10** of the electric motor **1**, the bracket **41** is disposed at an end in the rotation axis C direction, that is, disposed on the opposite output side of the shaft **32**. The non-magnetic portion **44** (end surface portion) of the bracket **41** includes a connection portion **45** connected to the first bearing storing portion **42** (see FIGS. **2**, **3**, and **5**). The non-magnetic portion **44** of the bracket **41** is integrally formed with the first bearing storing portion **42**, which is a magnetic portion, by insert molding. The non-magnetic portion **44** (end surface portion) is connected to the first bearing storing portion **42** (bearing house portion) at the connection portion **45**.

(23) The bracket **41** is attached to the end portion of the motor shell **10** on the opposite output side by using screws to serve as a lid for covering the opening of the motor shell **10** (main body). Note that the opening of the motor shell **10** may be provided toward the output side. In this case, the bracket **41** is disposed not on the opposite output side of the shaft **32** but on the output side of the shaft **32**.

(24) The non-magnetic portion **44** (end surface portion) of the bracket **41** is formed into a substantially circular plate shape having the outer shape in the radial direction, which expands to the outer circumferential surface of the motor shell **10** in the radial direction. The non-magnetic portion **44** of the bracket **41** forms a resin shell of the electric motor **1** together with the motor shell **10**. Additionally, the non-magnetic portion **44** includes protrusions **410**, which protrude outward in the radial direction relative to the outer circumferential surface of the motor shell **10**, as viewed from the rotation axis C direction. The protrusions **410** each abut on the basal end portion of a guard portion **102** of the motor shell **10**. The guard portion **102** will be described later.

(25) The protrusions **410** provided to the bracket **41** are formed as many as the guard portions **102** provided to the motor shell **10** (three positions). For example, the protrusions **410** are each formed into a trapezoid as viewed from the rotation axis C direction and each include, at the center portion thereof, a screw through hole **413** penetrating in the rotation axis C direction.

(26) Note that the bracket **41** includes a slitted groove **416** for providing an electrically conductive member **5** for measures against electrolytic corrosion, which will be described later, in the outer surface exposed to the outside in the electric motor **1** after assembling (see FIGS. **1** and **3**).

(27) The slitted groove **416** extends outward in the radial direction from the center portion of the bracket **41** (tubular connection portion **45** of the non-magnetic portion **44** to be described later) to the outer circumferential surface of the bracket **41**, and further extends in the axis direction from there to the position abutting on the motor shell **10**.

(28) The bracket **41** is fitted into the motor shell **10** and then screwed at screw holes **103** (to be described later) of the guard portions **102** of the motor shell **10** via the screw through holes **413** (see FIG. **1**).

(29) Further, the first bearing storing portion (bearing house portion) **42** for storing the first bearing **33** on the inner side (opposite output side) of the electric motor **1** is disposed at the center portion of the circular plate shape bracket **41**. The first bearing storing portion **42** is formed into, for example, a substantially bottomed cylindrical shape by press working.

(30) The second bearing storing portion (bearing house portion) **43** for storing the second bearing **34** on the inner side (output side) of the electric motor **1** is disposed at the center portion of the output side end portion of the motor shell **10**. The second bearing storing portion **43** is formed into, for example, a substantially bottomed cylindrical shape similarly to the first bearing storing portion **42**. The second bearing storing portion **43** is disposed inward (inner diameter side) relative to the annular permanent magnet portion **31** in the radial direction of the rotor **3**.

(31) The end surface portion **13** of the motor shell **10** includes a connection portion **14** that is connected to a flange portion **432** (to be described later) of the second bearing storing portion **43**.

(32) As shown in FIGS. **2** and **5**, the first bearing storing portion **42**, which is one of the pair of bearing house portions, includes a tubular portion **421** that holds the outer race side of the first bearing **33** from the radial direction, an annular flange portion **422** that extends outward in the radial direction of the rotor **3** from one end portion of the tubular portion **421** in the rotation axis C direction, and a coronal portion **423** that extends inward in the radial direction from the other end portion of the tubular portion **421** in the rotation axis C direction. The coronal portion **423** covers the other end side of the first bearing **33** in the rotation axis C direction. The outer circumferential edge of the annular flange portion **422** is located inward (inner circumferential side) in the radial direction of the rotor **3** relative to the permanent magnet portion **31**. In other words, the first bearing storing portion **42** is formed so as not to overlap with the permanent magnet portion **31** as viewed from the rotation axis C direction of the rotor **3**.

(33) Specifically, the first bearing storing portion **42** (bearing house portion) is disposed inward (inner diameter side) in the radial direction of the rotor **3** relative to the permanent magnet portion **31**, as viewed from the rotation axis C direction. Further, the outer circumferential edge portion of the flange portion **422** of the first bearing storing portion **42** (edge portion of the first bearing storing portion **42** on the outer diameter side) is covered with resin that is a non-magnetic substance. Specifically, in the bracket **41**, the outer circumferential edge portion of the flange portion **422** of the first bearing storing portion **42** is covered with the non-magnetic portion **44** made of resin.

(34) As described above, the bracket **41** is formed by the first bearing storing portion (magnetic portion) **42**, which is one of the pair of bearing storing portions (bearing house portions), and the non-magnetic portion **44** (end surface portion). The first bearing storing portion (magnetic portion) **42** is disposed on the inner diameter side relative to the permanent magnet portion **31** in the radial direction of the rotor **3**, and thus can prevent the flange portion **422** of the first bearing storing portion **42** serving as a magnetic portion from facing the permanent magnet portion **31** in the rotation axis C direction.

(35) This makes it possible to suppress a leakage flux flowing from the permanent magnet portion **31** to the first bearing storing portion (magnetic portion, bearing house portion) **42**. Furthermore, in the first bearing storing portion **42**, the outer circumferential edge portion of the flange portion **422**, which is disposed close to the permanent magnet portion **31** of the rotor **3**, is covered with the non-magnetic portion **44**. This makes it possible to block the path of the leakage flux flowing from the permanent magnet portion **31** to the first bearing storing portion (magnetic portion, bearing house portion) **42** formed of a magnetic substance by the non-magnetic portion (end surface portion) **44** formed of a non-magnetic substance, and thus further possible to suppress the leakage flux flowing from the permanent magnet portion **31** to the first bearing storing portion **42**.

(36) Note that such a structure for suppressing the leakage flux can be applied to not only the first bearing storing portion **42** side but also the second bearing storing portion **43** side. At that time, the second bearing storing portion **43** is formed into the shape similar to that of the first bearing storing portion **42** and includes a tubular portion **431** that holds the outer race side of the second bearing **34** from the radial direction, an annular flange portion **432** that extends outward in the radial direction of the rotor **3** from one end portion of the tubular portion **431** in the rotation axis C direction, and a coronal portion **433** that extends inward in the radial direction from the other end portion of the tubular portion **431** in the rotation axis C direction. Additionally, the second bearing storing portion **43** is disposed on the inner diameter side relative to the permanent magnet portion **31** in the radial direction of the rotor **3**.

(37) Further, the outer circumferential edge portion of the flange portion **432** of the second bearing storing portion **43** is covered with the end surface portion **13** (connection portion **14**) of the resin motor shell **10** that is a non-magnetic substance. This makes it possible to suppress the leakage flux flowing from the permanent magnet portion **31** to the second bearing storing portion **43**. The configuration for suppressing such a leakage flux may be applied to either one of or both of the first bearing storing portion **42** and the second bearing storing portion **43**, which are the pair of bearing house portions.

(38) The non-magnetic portion (end surface portion) **44** of the bracket **41** includes the connection portion **45** connected to the first bearing storing portion (bearing house portion) **42**. The connection portion **45** is formed into a substantially tubular shape, and the flange portion **422** of the bearing house portion (first bearing storing portion **42**) formed of a magnetic substance is inserted into and fixed to the side surface of the tubular connection portion **45** on the inner diameter side. Here, the tubular portion **421** of the first bearing storing portion **42** is not in contact with the non-magnetic portion **44** of the bracket **41** (is not covered with the non-magnetic portion **44**), and only the outer circumferential edge portion of the flange portion **422** is joined (connected) to the connection portion **45** of the non-magnetic portion (end surface portion) **44** so as to be covered therewith.

Further, a clearance portion (air gap) AG1 is formed between the tubular portion 421 of the first bearing storing portion 42 (bearing house portion) and the tubular connection portion 45 of the non-magnetic portion (end surface portion) 44 forming the resin shell.

(39) With this configuration, the deformation of the resin shell (motor shell 10, end surface portion (non-magnetic portion) 44) due to heat, shock, or the like hardly affects the first bearing 33.

Furthermore, the contact area between the connection portion 45 of the bracket 41 and the flange portion 422 of the first bearing storing portion 42 can be reduced, and thus the heat generated at the winding wound in the stator core 21 can be prevented from being transmitted to the first bearing 33 via the bracket 41. This makes it possible to suppress an increase in temperature of the first bearing 33 and prevent the first bearing 33 from deteriorating.

(40) Here, regarding the first bearing storing portion 42 side (opposite output side), the first bearing storing portion 42 being one of the pair of bearing storing portions, the annular portion 12 of the motor shell 10, which is integral with the stator 2, and the end surface portion (resin non-magnetic portion 44 of bracket 41) connected to the end portion of the annular portion 12 and expanding inward (inner circumferential side) in the radial direction are separately formed to serve as the resin shell of the motor 1. However, the annular portion 12 and the end surface portion in the resin shell may be integrally formed. In this embodiment, on the second bearing storing portion 43 side, the second bearing storing portion 43 being the other one of the pair of bearing storing portions, the annular portion 12 and the end surface portion are integrally formed as the resin shell.

(41) Specifically, on the second bearing storing portion 43 side (output side), the second bearing storing portion 43 being the other one of the pair of bearing storing portions, the second bearing storing portion 43 is formed into a shape similar to that of the first bearing storing portion 42. Note that the structures of the first bearing storing portion 42 and the second bearing storing portion 43 may be communalized, and the first bearing storing portion 42 and the second bearing storing portion 43 may be disposed upside down to communalize their constituent parts. The motor shell 10 is formed into a substantially bottomed cylindrical shape and includes the annular portion 12 of the motor shell 10, which is integral with the stator 2, and the end surface portion 13 of the motor shell 10, which is connected to the end portion of the annular portion 12 in the rotation axis C direction and expands inward (inner circumferential side) in the radial direction. Additionally, the end surface portion 13 of the motor shell 10 includes the cylindrical connection portion 14 connected to the second bearing storing portion 43. Further, similarly to the first bearing storing portion 42, the second bearing storing portion 43 also includes the flange portion 432 extending outward in the radial direction from the tubular portion 431, and only the outer circumferential edge portion of the flange portion 432 is inserted into and fixed to the side surface of the connection portion 14 of the resin shell (motor shell 10) on the inner diameter side. Further, a clearance portion (air gap) AG2 is formed between the tubular portion 431 of the second bearing storing portion 43 and the connection portion 14 of the resin shell (motor shell 10).

(42) With this configuration, the deformation of the resin shell (annular portion 12 and end surface portion 13 of motor shell 10) due to heat, shock, or the like hardly affects the second bearing 34.

Furthermore, the contact area between the connection portion 14 of the motor shell 10 and the flange portion 432 of the second bearing storing portion 43 can be reduced, and thus the heat generated at the winding wound in the stator core 21 can be prevented from being transmitted to the second bearing 34 via the resin shell. This makes it possible to suppress an increase in temperature of the second bearing 34 and prevent the second bearing 34 from deteriorating.

(43) As described above, the rotor 3 includes the coupling portion 35, to which the shaft 32 is fixed and which couples the permanent magnet portion 31 and the shaft 32 to each other. The permanent magnet portion 31 is disposed so as to face the cylindrical stator core 21 in the radial direction. The coupling portion 35 is disposed on the inner diameter side of the permanent magnet portion 31 annularly disposed. As shown in FIGS. 2 and 4, the coupling portion 35 includes an annular recess 36 that is recessed in the axis direction of the rotation axis C (rotation axis C direction). The recess

36 is formed such that the thickness of the coupling portion **35** in the rotation axis C direction at the position at which the recess **36** is formed is smaller than the thickness of the permanent magnet portion **31** in the rotation axis C direction. Additionally, the flange portion **422** of the first bearing storing portion **42** is disposed so as to overlap with the recess **36**, which is formed in the coupling portion **35**, in the rotation axis C direction. This makes it possible to form the annular recess **36** recessed toward the rotation axis C direction in the rotor **3**, so that the flange portion **422** of the first bearing storing portion **42** can be disposed within the recess **36**.

(44) In such a manner, a part of the first bearing storing portion **42** (flange portion **422**) enters the annular recess **36** recessed in the axis direction of the rotation axis C (rotation axis C direction), thus reducing the thickness of the electric motor **1** in the rotation axis C direction and downsizing the electric motor **1** in the rotation axis C direction.

(45) As shown in FIG. **4**, terminal pins **26** electrically connected to the winding (not shown) of the stator core **21**, and bosses **27** each serving as a guide used when a substrate (not shown) is attached are provided at the end portion (upper end portion in FIG. **4**) of the stator **2** on the opposite output side in the rotation axis C direction.

(46) The bracket **41** functions as an insulation cover for preventing the terminal pins **26** from being exposed to the outside of the electric motor **1**. In this embodiment, the terminal pins **26** are provided at three positions, and the bracket **41** is attached to the motor shell **10** so as to cover up those three positions.

(47) The bracket **41** includes a cover main body **414** to be attached along the upper end surface of the stator **2** and a fitting portion **415** integrally formed with the cover main body **414**. The cover main body **414** and the fitting portion **415** correspond to the non-magnetic portion **44** (end surface portion).

(48) The cover main body **414** is formed into a substantially circular plate shape as a whole. As shown in FIG. **3**, the fitting portion **415** is formed as an annular projection disposed at the outer circumferential edge portion of the cover main body **414**. The fitting portion **415** is fitted into the end portion of the motor shell **10** on the opposite output side (upper end surface of the motor shell **10** in FIG. **4**) from the rotation axis C direction, so that the motor shell **10** (main body) and the bracket **41** are aligned with each other, and the first bearing **33** is stored in the first bearing storing portion **42** provided to the bracket **41** as shown in FIG. **2**.

(49) The motor shell **10** includes the three guard portions **102** arranged at regular intervals in the circumferential direction at the end portion of the rotation axis C on the opposite output side. Note that any number of guard portions **102**, such as two or six guard portions **102**, may be provided, and the plurality of guard portions **102** need not be arranged at regular intervals. Those three guard portions **102** each protrude into a trapezoidal shape in the radial direction of the stator **2** (electric motor **1**) and each have a predetermined thickness in the rotation axis C direction.

(50) As shown in FIGS. **1** and **4**, each guard portion **102** includes a cutout portion **104** for fitting a vibration proof rubber bush **6** therein, the cutout portion **104** being formed from the outside in the radial direction of the stator **2** (electric motor **1**) to the inner radial direction. This cutout portion **104** is formed so as to connect a hole formed in each guard portion **102** to penetrate in the rotation axis C direction and the outer circumferential edge of the guard portion **102** to each other. Furthermore, each guard portion **102** includes the screw hole **103**, through which the above-mentioned bracket **41** is screwed.

(51) The lower surface (surface on the output side) of each guard portion **102** includes a recess portion **106** formed to easily hold the vibration proof rubber bush **6** (see FIGS. **1** and **4**).

(52) As shown in FIG. **4**, any one of the three guard portions **102** includes a slitted groove **105** for arranging the electrically conductive member **5** for measures against electrolytic corrosion (see FIG. **5**) along the rotation axis C direction, the slitted groove **105** being formed at the position on the most inner diameter side of the cutout portion **104** in the radial direction of the stator **2** (electric motor **1**). Along this slitted groove **105**, a slitted groove for the electrically conductive member **5** is

also formed on the side surface and the end surface portion **13** (surface on the output side) of the motor shell **10** so as to extend in the axis direction of the rotation axis C and the radial direction (not shown).

(53) The electrically conductive member **5** is a strip-shaped member for electrical conduction between the first bearing **33** and the second bearing **34**. The electrically conductive member **5** is formed by, for example, punching a steel plate into a strip shape and bending the obtained steel plate into a substantially U shape (squared U shape). The electrically conductive member **5** is disposed along the outer surfaces of the motor shell **10** and the bracket **41** (see FIG. 5). The electrically conductive member **5** allows the potentials of the first bearing **33** and the second bearing **34** on the outer race side to be the same, and thus the generation of electrolytic corrosion on the bearings can be suppressed.

(54) Here, when the bracket **41** is fitted into the motor shell **10**, the slitted groove **416** of the bracket **41** and the slitted groove **105** formed in the outer surface of the motor shell **10** become continuous, and both the slitted grooves become a guide into which the strip-shaped electrically conductive member **5** is to be embedded. This makes it possible to prevent the strip-shaped electrically conductive member **5** from protruding from the surface of the shell of the permanent magnet electric motor **1** and from dropping. As shown in FIG. 5, the electrically conductive member **5** is disposed to extend from the position of the flange portion **422** of the first bearing storing portion **42** to the position of the flange portion **432** of the second bearing storing portion **43** through the slitted groove **416** of the bracket **41**, the slitted groove **105** of the guard portion **102**, and the slitted groove of the outer surface of the motor shell **10** (not shown).

(55) As described above, in the electric motor **1**, the two bearings of the first bearing **33** and the second bearing **34** are disposed, and the two bearing house portions of the first bearing storing portion **42** and the second bearing storing portion **43** are disposed. The bearing house portions **42** and **43** store the bearings **33** and **34**, respectively. Further, the electric motor **1** includes the electrically conductive member **5** for electrical conduction between the two bearings **33** and **34**. Additionally, at least one end portion (both end portions in the embodiment) of the electrically conductive member **5** is disposed in the clearance portion AG1, AG2 formed between the tubular portion **421**, **431** of the bearing house portion **42**, **43** and the connection portion **45**, **14** of the resin shell (end surface portion **44**, **13**) (see FIG. 5). With this configuration, the end portion of the electrically conductive member **5** for preventing electrolytic corrosion is disposed in the clearance portion AG1, AG2 that is a tubular clearance, and thus it is possible to prevent the electrically conductive member **5** from dropping when the end portion of the electrically conductive member **5** is touched by a human hand to thereby hinder the electrical conduction between the bearings. Further, the end portion of the electrically conductive member **5** can be press-fitted into the clearance portion AG1, AG2 described above to be easily fixed to the outer circumferential surface of the bearing house portion **42**, **43**.

(56) Further, before the vibrationproof rubber bush **6** is fitted into the guard portion **102**, the electrically conductive member **5** is inserted into the slitted groove **105** in advance, so that the vibrationproof rubber bush **6** can press the electrically conductive member **5** from the outside, and the electrically conductive member **5** can be prevented from dropping.

(57) As described above, in this embodiment, since the resin shell (motor shell **10**, non-magnetic portion **44**) having a large amount of deformation due to heat does not cover the entire bearing house portions **42** and **43**, the deformation of the first and second bearings **33** and **34** and the bearing house portions **42** and **43** along with the deformation of the resin shell can be suppressed. Further, the heat generated at the winding wound in the stator core **21** can be prevented from being transmitted to the bearings **33** and **34** via the resin shell.

(58) Meanwhile, the resin shell (connection portion **14** of motor shell **10**, connection portion **45** of bracket **41**) is connected to the outer edge of the flange portion **422**, **432** of the bearing house portion **42**, **43**, so that the bearing house portion **42**, **43** can be fixed to the resin shell.

(59) Further, it is possible to prevent the pair of bearings (first bearing **33** and second bearing **34**) and the pair of bearing house portions (first bearing storing portion **42**, second bearing storing portion **43**) from protruding in the axis direction of the rotation axis C and to downsize the electric motor **1** in the direction of the rotation axis.

REFERENCE SIGNS LIST

(60) **1** electric motor **10** motor shell (resin shell) **12** annular portion **13** end surface portion **14** connection portion **2** stator **21** stator core **3** rotor **31** permanent magnet portion **32** shaft **33** first bearing **34** second bearing **35** coupling portion **36** recess **41** bracket **42, 43** bearing storing portion (bearing house portion) **421, 431** tubular portion **422, 432** flange portion **423, 433** coronal portion **44** end surface portion (non-magnetic portion) **45** connection portion **5** electrically conductive member **AG1, AG2** clearance portion **C** rotation axis

Claims

1. An electric motor, comprising: a columnar rotor; a shaft disposed along a rotation axis of the rotor; a cylindrical stator core disposed on an outer circumferential side of the rotor; a resin shell that covers the stator core; a bearing that rotatably supports the shaft; and a bearing house portion that stores the bearing, wherein the resin shell includes an annular portion integral with the stator core, and an end surface portion connected to an end portion of the annular portion and expanding from the annular portion to an inner circumferential side, the end surface portion includes a connection portion to be connected to the bearing house portion, the bearing house portion includes a tubular portion and a flange portion extending from an end of the tubular portion to the outer circumferential side, an outer circumferential edge portion of the flange portion is fixed to the connection portion of the resin shell, and the rotor includes an annular permanent magnet portion, a coupling portion that is disposed on the inner diameter side of the permanent magnet portion and couples the permanent magnet portion and the shaft to each other, and an annular recess that is recessed in a center of an axis direction of the rotation axis, and the permanent magnet portion and the coupling portion face each other with the recess therebetween in a radial direction perpendicular to the rotation axis.

2. The electric motor according to claim 1, wherein the tubular portion of the bearing house portion and the connection portion of the resin shell form a clearance portion therebetween.

3. The electric motor according to claim 2, wherein the bearing has a first bearing provided at one end of the shaft and a second bearing provided at an other end of the shaft, the bearing house portion has a first bearing house portion that stores the first bearing and a second bearing house portion that stores the second bearing, the electric motor further includes an electrically conductive member for electrical conduction between the first bearing and the second bearing, and at least one end portion of the electrically conductive member is disposed in the clearance portion.

4. The electric motor according to claim 1, wherein the connection portion of the resin shell is formed into a tubular shape extending so as to protrude from an end portion on an inner circumferential side of the end surface portion toward a center in an axis direction of the rotation axis, and the flange portion of the bearing house portion is inserted into and fixed to a side surface of the connection portion on an inner diameter side.

5. An electric motor, comprising: a columnar rotor; a shaft disposed along a rotation axis of the rotor; a cylindrical stator core disposed on an outer circumferential side of the rotor; a resin shell that covers the stator core; a bearing that rotatably supports the shaft; and a bearing house portion that stores the bearing, wherein the resin shell includes an annular portion integral with the stator core, and an end surface portion connected to an end portion of the annular portion and expanding from the annular portion to an inner circumferential side, the end surface portion includes a connection portion to be connected to the bearing house portion, the bearing house portion includes a tubular portion and a flange portion extending from an end of the tubular portion to the outer

circumferential side, an outer circumferential edge portion of the flange portion is fixed to the connection portion of the resin shell, the rotor includes an annular permanent magnet portion, a coupling portion that is disposed on the inner diameter side of the permanent magnet portion and couples the permanent magnet portion and the shaft to each other, and an annular recess that is recessed in a center of an axis direction of the rotation axis, and the permanent magnet portion and the coupling portion face each other with the recess therebetween in a radial direction perpendicular to the rotation axis.

6. The electric motor according to claim 5, wherein the tubular portion of the bearing house portion and the connection portion of the resin shell form a clearance portion therebetween.

7. The electric motor according to claim 6, wherein the bearing has a first bearing provided at one end of the shaft and a second bearing provided at an other end of the shaft, the bearing house portion has a first bearing house portion that stores the first bearing and a second bearing house portion that stores the second bearing, the electric motor further includes an electrically conductive member for electrical conduction between the first bearing and the second bearing, and at least one end portion of the electrically conductive member is disposed in the clearance portion.

8. The electric motor according to claim 5, wherein the connection portion of the resin shell is formed into a tubular shape extending so as to protrude from an end portion on an inner circumferential side of the end surface portion toward a center in an axis direction of the rotation axis, and the flange portion of the bearing house portion is inserted into and fixed to a side surface of the connection portion on an inner diameter side.

9. An electric motor, comprising: a columnar rotor; a shaft disposed along a rotation axis of the rotor; a cylindrical stator core disposed on an outer circumferential side of the rotor; a resin shell that covers the stator core; a bearing that rotatably supports the shaft; and a bearing house portion that stores the bearing, wherein the resin shell includes an annular portion integral with the stator core, and an end surface portion connected to an end portion of the annular portion and expanding from the annular portion to an inner circumferential side, the end surface portion includes a connection portion to be connected to the bearing house portion, the bearing house portion includes a tubular portion and a flange portion extending from an end of the tubular portion to the outer circumferential side, an outer circumferential edge portion of the flange portion is fixed to the connection portion of the resin shell, the rotor includes an annular recess that is recessed in a center of the axis direction of rotation axis, the flange portion of the bearing house portion is disposed within the recess, and the tubular portion of the bearing house portion and the connection portion of the resin shell form a clearance portion therebetween.
